Using Machine Learning (XGBoost) Methods to Analyze Traffic Flows in Lima, Peru

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As the global population grows exponentially, an increasing proportion of the world has begun to live in urban environments, with an estimated 68% predicted to live in cities by 2050[13]. Accurate traffic flow prediction models are necessary to optimize traffic control systems and inform sustainable, safe, and cost-effective infrastructure. Recent advances in machine learning (ML) and deep learning techniques enhance Intelligent Transportation Systems (ITS) by more accurately predicting traffic flows than historical averages or linear regression techniques[11]. However, the majority of studies concentrate on highly-developed cities where data is readily available, while fewer studies focus on the implementation of ML-powered traffic control monitoring systems in developing cities [8]. Since developing cities are likely to suffer from uneven development which begets traffic congestion and traffic-related accidents, understanding traffic flows in developing cities is crucial for policymakers and urban planners.

This project aims to create a dynamic traffic flow model of Lima, Perú, utilizing Extreme Gradient Boosting (XGBoost) and Random Forest (RF) machine learning techniques. XGBoost and RF were chosen due to their relatively low computational requirement compared to deep learning neural networks, their high performance with limited data, and their proficiency with tabular data. Open-Source Map (OSM) data will be used to generate a road map of the city, while additional open-source data from the Autoridad de Transporte Urbano (ATU), Indicadores Mensuales de Tráfico, and TomTom will be used to model population flows according to spaciotemporal indicators. Feature importance analysis will be applied to determine the impact of each feature on the traffic outcome. The end product will be a model created using GIS software that can predict traffic flows given certain conditions, such as time, date, and weather. The model will additionally be able to generate predictive outcomes of road closures or the construction of new routes. A potential GUI for the software is displayed in Figure 1.

The project is informed by a variety of machine learning research initiatives which applied machine learning techniques to analyze traffic in urban environments across the globe. A variety of studies have displayed the effectiveness of XGBoost and RF in modeling traffic flows in developing countries compared to ANN, LSTM, or SVM deep learning models [8][1][5][10]. Feature importance analyses are applied in Abdulrashid et al., [1] and in Dorosan et al., [8], which use a variety of Shapley's additive explanation (SHAP), PFI, and FIRM scores to compare the behvaiors of various models. The GIS mapping aspect of the project is inspired by a study in Ali Menjeli, Algeria, which combines LSTM-analyzed data to create isochromes that are mapped using GIS software [9].

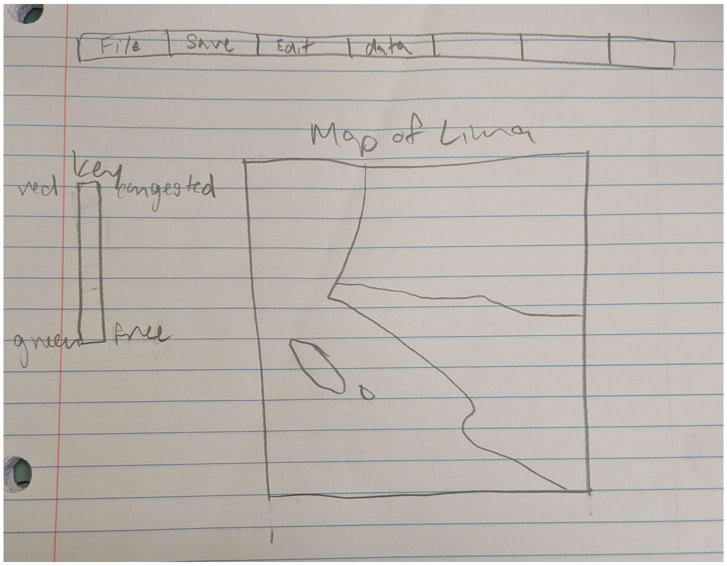

Figure 1: A hand-drawn GUI of the final traffic-mapping software.

Appendix

Product features.

- Creation of pre-processed data tables (two weeks)
- Visualization of the statistical baseline (found using historical average linear regression techniques) (one week)
- Table of XGBoost results. (one week)
- Table of RF results. (one week)
- Plot to display the importance of each feature on the outcome (calculated using SHAP, PFI, and/or FIRM). (one week)
- Loss plot to determine the accuracy of each model. (one week)
- Map results to GIS to create a color-coded visualization of congestion on each street. (two weeks)
- Hypothetical: Create software that will generate color-coded maps of Lima based on certain parameters. (three weeks)
- Implement capacity of GIS model to vary depending on time, date, and weather features. (two weeks)
- Implement capacity of GIS model to respond to conditions such as a closed road or or an additional freeway. (one week)

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